



Bayes' Theorem

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Conditional Probability — The Foundation

Conditional Probability $P(A|B)$ measures the probability of event A occurring, given that event B has already occurred.

Definition:

$$P(A | B) = P(A \cap B) / P(B) \quad \text{provided } P(B) > 0$$

Intuitive Example

A bag has: 3 Red balls, 2 Blue balls

Event A = 'draw a Red ball'

Event B = 'draw a ball on Monday'

$$P(A) = 3/5 = 0.60$$

$$P(A|B) = 3/5 = 0.60 \quad (\text{Monday doesn't help here})$$

But if B = 'a Red was drawn first (without replacement)':

$$P(A|B) = 2/4 = 0.50 \quad (\text{reduced pool!!})$$

Key Properties

Multiplication Rule:

$$P(A \cap B) = P(A|B) \times P(B) = P(B|A) \times P(A)$$

Independence:

A and B are independent iff

$$P(A|B) = P(A) \quad \leftarrow \quad B \text{ gives no information about } A$$

Law of Total Probability:

$$P(B) = P(B|A) \cdot P(A) + P(B|\neg A) \cdot P(\neg A)$$

Bayes' Theorem

Bayes' Theorem (Thomas Bayes, 1763)

$$P(A | B) = P(B | A) \times P(A) / P(B)$$

Expanded form (using Law of Total Probability):

$$P(A|B) = P(B|A) \times P(A) / [P(B|A) \times P(A) + P(B|\neg A) \times P(\neg A)]$$

P(A) = Prior Probability

Our INITIAL probability estimate for A BEFORE observing B. What we believe about A without any new evidence.

P(B|A) = Likelihood

The probability of observing evidence B IF A is true. How likely is the evidence under our hypothesis?

P(A|B) = Posterior Probability

The UPDATED probability of A AFTER observing evidence B. This is what Bayes' Theorem computes for us.

P(B) = Marginal Probability

Total probability of observing B regardless of A. Acts as a normalizing constant (denominator).

Classic Example: Medical Diagnostic Test

Scenario: A rare disease D affects 1% of the population. A test T has 99% accuracy (both sensitivity and specificity). If a person tests positive, what is the probability they actually have the disease?

$$P(D) = 0.01$$

Prior probability of having the disease (1% of population)

$$P(\neg D) = 0.99$$

Prior probability of NOT having the disease (99%)

$$P(T^+ | D) = 0.99$$

Sensitivity: Test correctly detects disease 99% of the time

$$P(T^+ | \neg D) = 0.01$$

False Positive Rate: Test incorrectly says positive 1% of the time for healthy people

Question: A patient tests positive (T^+). What is $P(D | T^+)$? Most people guess $\approx 99\%$... the answer is SHOCKING.

Solving with Bayes' Theorem

Apply the expanded form of Bayes' Theorem:

Formula: $P(D|T^+) = P(T^+|D) \times P(D) / [P(T^+|D) \times P(D) + P(T^+|\neg D) \times P(\neg D)]$

Substitute values: $P(D|T^+) = (0.99 \times 0.01) / [(0.99 \times 0.01) + (0.01 \times 0.99)]$

Numerator: $0.99 \times 0.01 = 0.0099$

Denominator: $(0.99 \times 0.01) + (0.01 \times 0.99) = 0.0099 + 0.0099 = 0.0198$

Final Answer: $P(D|T^+) = 0.0099 / 0.0198 = 0.50 = 50\% !!$

Why only 50%? Of 10,000 people: 100 have disease (99 test positive) + 9,900 healthy (99 test false-positive). So of 198 positive tests, only 99 truly have the disease = 50%!

Key Lesson: Low prior probability dramatically reduces the posterior even with accurate tests.

Applications in Computer Science

Bayes' Theorem is the foundation of probabilistic reasoning in AI and data science:

Naive Bayes Spam Filter

Given an email, compute $P(\text{spam} \mid \text{words in email})$. Train on labeled emails to learn $P(\text{word} \mid \text{spam})$ and $P(\text{word} \mid \text{not-spam})$. The 'Naive' assumption: words are independent given the class. Very fast and still competitive.

Medical Diagnosis AI

AI systems compute $P(\text{disease} \mid \text{symptoms, test results})$. Updated with new evidence as more tests are run. IBM Watson Health uses Bayesian reasoning to recommend treatments based on patient history and literature.

Search & Recommendation

Google's ranking algorithms incorporate Bayesian principles. Collaborative filtering in recommendation systems (Netflix, Amazon) estimates $P(\text{user likes item} \mid \text{ratings of similar users})$ using Bayesian updates.

Intrusion Detection

Network security systems compute $P(\text{attack} \mid \text{observed traffic patterns})$. Start with $P(\text{attack})$ as the base rate, update with each suspicious packet. Bayesian Networks model complex multi-step attack scenarios.



THANK YOU

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